# Clay Emerson Wood, Ph.D.

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## Research Profile

Experimental geoscientist and physicist who designs and builds the high-pressure laboratory systems used to measure how rock fractures, flows, and reacts under reservoir conditions. Produced the first experimental evidence of reaction-driven fracturing in ultramafic rock during $CO_2$ mineralization and established quantitative links between fracture structure and the hydraulic and elastodynamic properties of fractured rock — work spanning custom instrumentation, synchrotron X-ray CT, ultrasonic sensing, and HPC-scale analysis. This combination of hands-on experimental capability and quantitative modeling translates directly to geothermal reservoir characterization, subsurface thermal energy storage, and well repurposing, where coupled thermal–hydraulic–mechanical–chemical behavior governs performance.

## Education

**Ph.D., Geosciences** 2023
Pennsylvania State University, State College, PA
Dissertation — Imaging elastodynamic and hydraulic properties of fractured rock: Exploring effects of dynamic stressing, shearing, fracture aperture, and roughness

**B.S., Physics** 2016
Emory University, Atlanta, GA

## Advisors

- **Post Doc advisors:** Prof. Nicola Tisato (The University of Texas at Austin) and Prof. Nicolas Espinoza (The University of Texas at Austin).
- **Ph.D. advisors:** Prof. Chris Marone (Pennsylvania State University & Dipartimento di Scienze della Terra, La Sapienza Università di Roma) and Prof. Parisa Shokouhi (Pennsylvania State University).
- **B.S. advisor:** Prof. Justin Burton (Emory University)

## Relevant Technical Expertise

- **Experimental & instrumentation:** high-pressure flow-through and true-triaxial apparatus design and operation; custom hardware (CAD, 3D printing); MATLAB instrument control and acquisition; ultrasonic and X-ray CT integration.
- **Rock physics & flow:** permeability evolution, poromechanics, reactive transport, hydro-mechanical coupling under reservoir conditions.
- **Computation:** Python, Julia, MATLAB; signal/image processing, inverse and reduced-order modeling, workflow automation; HPC (TACC).
- **Characterization:** CT-based fracture analysis, elastic-wave/ultrasonic measurement, quantitative data interpretation.
- **Leadership:** multi-institution programs, external user facilities, graduate & undergraduate mentoring, peer-reviewed dissemination.

## Professional Appointments

**Geophysics Subject Matter Expert** 03/2026 – Present
*Handshake AI (Contract)*

- Design evaluation questions, grading rubrics, and reference solutions used to benchmark and improve the scientific reasoning of frontier large language models in geosciences and physics.
- Serve as domain expert grading model responses for technical accuracy across subsurface, rock-physics,

and geophysics problems.

**Postdoctoral Fellow – Reaction-Driven Cracking and Rock Characterization** 09/2023 – 12/2025
*University of Texas at Austin, Austin, TX*

- Designed and built high-pressure flow-through experiments on ultramafic rock (confining pressure up to 5 MPa, 100°C), resolving reaction-driven fracture propagation and permeability evolution across month-long reactive-flow experiments.
- Engineered integrated workflows coupling hydraulic systems, pressure/flow sensing, and X-ray CT to resolve transport and mechanical change in reacting rock.
- Built Python, Julia, and MATLAB pipelines for analysis of TBs of CT imaging data, permeability interpretation, and cross-experiment comparison.
- Co-advised one M.S. and one B.S. student on experimental rock mechanics research.

**Graduate Researcher – Rock Mechanics Laboratory** 07/2017 – 08/2023
*Pennsylvania State University, State College, PA*

- Designed and operated true-triaxial and flow-through experiments measuring permeability, elastic response, and hydro-mechanical coupling under controlled stress up to 20 MPa and pore pressure up to 5 MPa.
- Established quantitative relationships linking fracture aperture, roughness, and microstructure to hydraulic and geophysical properties, published across multiple *JGR: Solid Earth* papers.
- Ran synchrotron X-ray CT and ultrasonic experiments at the GeoSoilEnviroCARS beamline, Argonne National Laboratory, and contributed to analysis characterizing fracture evolution and transport.

**Research Assistant – Fluid and Granular Mechanics Laboratory** 06/2016 – 07/2017
*Emory University, Atlanta, GA*

- Ran precision experiments on fluid transport, wetting, and interfacial dynamics in multiphase systems, contributing to publications in *Physical Review Letters* and *Physical Review Fluids*.

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## Publications

11 peer-reviewed publications and 6 manuscripts in review/preparation; first author on six, in journals including Physical Review Letters, Geophysical Research Letters, and JGR: Solid Earth.

**Google Scholar: 215 citations, h-index 7**

**C. Wood**, N. Tisato, N. Espinoza, E. Ukar. (2026) Reaction-driven cracking and permeability evolution in ultramafic rock via $CO_2$ mineralization: Impacts of precipitation, dissolution, and fracture closure. *In Prep.*

**C. Wood**, A. Kenigsberg, J. Rivière, C. Marone, D. Saffer. (2026) Evolution of Permeability During Shear of Fault Gouge: Effects of Composition and Fabric. *In Prep.*

**C. Wood**, N. Espinoza, N. Tisato, E. Ukar. (2026) Experimental evidence of reaction-driven fracturing in ultramafic rock driven by $CO_2$ mineralization. *Geophys. Res. Lett., Under Rev.*

S. Almeshari, **C. Wood**, E. Ukar, A. Fall, N. Tisato. (2026) Carbon Mineralization in Ultramafic Rocks: Insights from Flow-Through CT-Imaged Experiments. *Geophys. Res. Lett., Under Rev.*

E. Bozek, P. Borate, Chun-Yu Ke, **C. Wood**, L. Pillarisetti, M. Rivers, D. Elsworth, J. Rivière, P. Shokouhi. (2026) Linear and nonlinear elastodynamic signatures of fractured rock in relation to fault characteristics inferred from in-situ synchrotron X-ray computed tomography. *JGR: Solid Earth, Under Rev.*

**C. Wood**, J. Rivière, D. Elsworth, C. Marone, P. Shokouhi. (2026) Poromechanics of particle remobilization and interface stiffness of dynamically stressed fractured rock. *JGR: Solid Earth, Under Rev.*

A. Eijsink, **C. Wood**, C. Marone, P. Shokouhi, J. Rivière, D. Elsworth. (2025) Evolution of frictional properties, permeability and elastic parameters due to surface roughness changes during fault maturation in laboratory shear experiments. *EPSL*

**C. Wood**, P. Manogharan, A. Rathbun, J. Rivière, D. Elsworth, C. Marone, P. Shokouhi. (2024) Relating fracture aperture to hydro-mechanical properties of dynamically-stressed tensile-fractured rock. *JGR: Solid Earth*

R. Affinito, **C. Wood**, S. Marty, D. Elsworth, C. Marone. (2024) The stability transition from stable to

unstable frictional slip with finite pore pressure. *Geophys. Res. Lett., e2023GL105568.*

**C. Wood**, P. Shokouhi, P. Manogharan, J. Rivière, D. Elsworth, C. Marone. (2021) Imaging Elastodynamic and Hydraulic Properties of In Situ Fractured Rock: An Experimental Investigation Exploring Effects of Dynamic Stressing and Shearing. *JGR: Solid Earth e2020JB021521.*

S. Shreedharan, M. Ikari, **C. Wood**, D. Saffer, L. Wallace, C. Marone. (2021) Frictional and Lithological Controls on Shallow Slow Slip at the Northern Hikurangi Margin. *G-Cubed.*

P. Manogharan, **C. Wood**, C. Marone, D. Elsworth, J. Rivière, P. Shokouhi. (2021) Nonlinear elastodynamic behavior of intact and fractured rock under in-situ stress and saturation conditions. *JMPS 153, 104491.*

P. Manogharan, **C. Wood**, C. Marone, D. Elsworth, J. Rivière, P. Shokouhi. (2021) Experimental Investigation of Elastodynamic Nonlinear Response of Dry Intact, Fractured and Saturated Rock. *Rock Mech Rock Eng, 125.*

A. Mouat, **C. Wood**, J. Pye, J. Burton. (2020) Tuning Contact Line Dynamics and Deposition Patterns in Volatile Liquid Mixtures. *Phys. Rev. Lett, 124, 064502*

P. Shokouhi, J. Jin, **C. Wood**, J. Rivière, B. Madara, D. Elsworth, C. Marone. (2020). Dynamic stressing of naturally fractured rocks: On the relation between transient changes in permeability and elastic wave velocity. *Geophys. Res. Lett., 47, e2019GL083557.*

A. Mouat, **C. Wood**, J. Pye, J. Burton. (2019) Liquid deposition through evaporation. *Phys. Rev. Fluids, 4, 100512.*

J. Pye, **C. Wood**, J. Burton. (2018) Precursors to Molecular Slip on Smooth Hydrophobic Surfaces. *Phys. Rev. Lett, 121, 134501.*

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## Selected Conference Talks & Presentations

**C. Wood**, N. Tisato, N. Espinoza, E. Ukar, A. Stokreef. Enhancing fracture permeability due to in-situ mineral carbonation: an experimental investigation into reaction-driven cracking in ultramafic rocks. Poster. American Geophysical Union Fall Meeting. December 2024.

**C. Wood**, C. Ke, J. Rivière, D. Elsworth, C. Marone, P. Shokouhi. Decoupling the poromechanics of particle remobilization and interface stiffness of dynamically stressed tensile fractured rock. Poster. American Geophysical Union Fall Meeting. December 2022.

**C. Wood**, C. Ke, A. Rathbun, J. Rivière, D. Elsworth, C. Marone, P. Shokouhi. Probing the micromechanical features of a fracture interface using a multi-physics approach: A numerical investigation relating asperity deformation with fluid flow. Talk. European Geophysical Union. May 2022.

**C. Wood**, P. Manogharan, J. Rivière, D. Elsworth, C. Marone, P. Shokouhi. Relating fracture aperture to hydraulic and elastodynamic properties of dynamically-stressed rock under true-triaxial stress conditions. Talk. American Geophysical Union Fall Meeting. Washington D.C., USA. December 2021.

**C. Wood**, P. Manogharan, J. Rivière, D. Elsworth, C. Marone, P. Shokouhi. Relating fracture aperture to hydraulic and elastodynamic properties of dynamically-stressed rock under true-triaxial stress conditions. Poster. The Physics of Earthquake Faulting. Rome, IT. September 2021.

**C. Wood**, P. Manogharan, S. Zi, J. Rivière, D. Elsworth, C. Marone, P. Shokouhi. The Relation Between Fracture Aperture and Hydro-mechanical Properties: An Experimental and Analytical Approach. Poster. American Geophysical Union Fall Meeting. Washington D.C., USA. December 2020.

**C. Wood**, P. Manogharan, J. Jin, J. Rivière, D. Elsworth, C. Marone, P. Shokouhi. The Influence of Fracture Aperture on Permeability and Elastic Nonlinearity: An Integrated Analytical and Empirical Investigation. Talk. American Geophysical Union Fall Meeting. Washington D.C. December 2019.

**C. Wood**, B. Madara, P. Shokouhi, J. Jin, J. Rivière, D. Elsworth, C. Marone. The Effect of Roughness on the Elasticity and Permeability of Fractured Media. Poster. Schatzalp 3rd Induced Seismicity Workshop. Davos, CH. March 2019.

**C. Wood**, B. Madara, P. Shokouhi, J. Jin, J. Rivière, D. Elsworth, C. Marone. The Effect of Roughness on the Elasticity and Permeability of Fractured Media. Poster. American Geophysical Union Fall Meeting. Washington D.C. December 2018.

**C. Wood**, B. Madara, P. Shokouhi, J. Jin, J. Rivière, D. Elsworth, C. Marone. The Effect of Roughness on the Elasticity and Permeability of Fractured Media. Poster. International School of Physics, Enrico

Fermi. Varenna, It. December 2018.

**C. Wood**, J. Pye, J. Burton. Liquid "Coffee Rings" and the Spreading of Volatile Liquid Mixtures. Talk. American Physical Society March Meeting. New Orleans, USA. March 2017

**C. Wood**, J. Pye, J. Burton. Quantifying stick-slip contact line motion of evaporating sessile droplets. Talk. American Physical Society March Meeting. Baltimore, USA. March 2016

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## Technical Project

**Local-First AI Productivity & Reference Manager** Personal Project

- Building a desktop application that uses local large language models and semantic indexing to organize notes and references by topic without manual tagging, surfacing cross-project connections through fully local inference to preserve privacy.

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## Teaching

- Geosc 597 Techniques in Geophysical Experimentation. Co-instructor with Dr. Chris Marone for laboratory skills course.
- Geosc 452: Hydrogeology. Teaching Assistant. Instructors: Dr. Byron Parizek, Dr. John Hooker

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## Professional Memberships

- American Geophysical Union member: 2018-Present
- American Physical Society member: 2016-Present

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## Honors & Awards

- Krynine Travel Grant, PSU Dept. of Geosciences 2018–2023
- Shell Geosciences Energy Research Facilitation Award, PSU Dept. of Geosciences 2019 & 2022
- Eagle Scout, Boy Scouts of America 2010

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